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Age-Related Changes in Blood Volume Pulse Wave at Fingers and Ears

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Abstract

Objective: The decline in vascular elasticity with aging can be manifested in the shape of pulse wave. The study investigated the pulse wave features that are sensitive to age and the pattern of these features change with increasing age were examined. **Methods:** Five features were proposed and extracted from the photoplethysmography (PPG)-based pulse wave or its first derivative wave. The correlation between these PPG features and ages was studied in 100 healthy subjects with a wide range of ages (20-71 years). Piecewise regression coefficients were calculated to examine the rates of change of the PPG features with age at different age stages. **Results:** The proposed PPG features obtained from the finger showed a strong and significant correlation with age (with $r = 0.76 - 0.77$, $p < 0.01$), indicating higher sensitivity to age changes compared to the PPG features reported in previous studies (with $r = 0.66 - 0.75$). The correlation remained significant even after correcting for other clinical variables. The rate of change of the PPG feature values was found to be significantly faster in subjects aged ≥ 40 years compared to those aged < 40 years in the healthy population. This rate of change was similar to the age-related progression of arterial stiffness evaluated by pulse wave velocity (PWV), which is considered a gold standard for evaluating vascular stiffness. **Conclusions:** The proposed PPG features showed a high correlation with chronological age in healthy subjects and exhibited a similar age-related change trend as PWV. **Significance:** With the convenience of PPG measures, the proposed age-related features have the potential to be used as biomarkers for vascular aging and estimating the risk of cardiovascular disease.

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Keywords

- Age-related features
- cardiovascular disease risk
- photoplethysmography (PPG)
- vascular aging

ASJC Scopus subject areas

- Computer Science Applications
- Health Informatics
- Electrical and Electronic Engineering
- Health Information Management

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KW - photoplethysmography (PPG)

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